GALSNP: A GALS Reconfigurable Neuromorphic Processor Enabling Manifold-Based Hybrid ANN-SNN and Gradient-Free On-Chip Adaptation

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Spiking neural network (SNN) based neuromorphic processors have been increasing high energy efficiency, but there still exist three major issues (Fig. 1) [1]-[4]. Firstly, sensor noises arising from environmental interference and operating condition variations induce feature distribution shifts, severely degrading inference accuracy [2]. Secondly, as the deployment of gradient backpropagation (BP) based on-chip adaptation often operates under zero-shot and noisy conditions, which lead to noise-induced gradient bias, still causing parameter drift and catastrophic forgetting [3]. Thirdly, under multi-layer data-dependent workloads, synchronous SNN designs incur redundant clock switching during inter-layer waiting, while asynchronous designs increase hardware overhead, scalability, and timing challenges, but achieve limited clock-power savings under dense computation due to continuous event triggering [4].

Inspired by neural manifold theory, which hypothesizes that neural population activity lies on a low-dimensional structure and tends to preserve class geometry under noise [1], our research finds that SNN population activity can exhibit similar manifold-like representations in spike-based feature spaces. Based on this insight, we proposes a GALSNP design with three distinct features (Fig. 2): 1) a novel manifold-based neuromorphic processor with a reconfigurable subspace projection engine for both hybrid ANN-SNN feature extraction and manifold-based class-specific energy evaluation, achieving robust inference accuracy under severe input noises; 2) a gradient-free on-chip adaptation technique based on manifold descriptor, improving zero-shot learning accuracy without catastrophic forgetting; 3) a GALS architecture that exploits both multi-layer data dependency in inference and manifold descriptor adaption in training, eliminating redundant clock switching and avoiding asynchronous event trigger to improve energy efficiency.

Figure 3 shows the proposed manifold-based hybrid ANN-SNN algorithm flow. Most existing neuromorphic processors [4]-[10] adopt conventional SNNs with point-wise decision boundaries, which cannot effectively distinguish noise-induced feature perturbations from subject-specific intra-class variations, thereby degrading classification robustness. To address this issue, we reformulate the classification task as a manifold energy minimization problem to suppress only the noise-induced off-manifold deviations while preserving intrinsic on-manifold subject variations. Specifically, in the ANN-SNN extracted latent feature space, each class is modeled as a class-specific manifold $\mathcal{M}$, which is represented by a subspace centered at the class mean μ_c and spanned by the dominant intra-class variation directions U_c. During inference, noisy data z_c is projected onto the tangent subspace $T_{\mu_c}\mathcal{M}$, where noise-induced perturbations are assumed as orthogonal deviations from the class-specific manifold. Classification is then performed through manifold energy E_c evaluation, which mainly measures the orthogonal distance from z_c to the tangent subspace, thereby achieving significantly higher robustness than the Euclidean-distance-based method. In addition, the hybrid ANN-SNN scheme leverages both the strong feature extraction capability of ANN and the high energy efficiency of SNN. The ANN-SNN computation, and manifold energy evaluation are mapped onto the reconfigurable sparsity-aligned processing cores (RSPC) in the proposed GALS processor.

Figure 3 also illustrates the proposed gradient-free manifold-descriptor-based on-chip adaptation technique and hardware architecture mapping. Instead of using gradient-based online training suffering catastrophic forgetting, our proposed adaptation technique confines the update of manifold descriptor (μ_c & U_c) to the local manifold subspace of new subject, thereby eliminating catastrophic forgetting without changing the subspace of old samples. This technique significantly reduces adaptation computation, while achieving high zero-shot classification accuracy. The major computation of manifold-based adaptation is mapped into the RSPC and postprocessing cores (PPC) in the proposed processor.

Figure 4 shows the sparsity-aware RSPC for the manifold-based hybrid ANN-SNN design. The RSPC reconfigure MAC units with multiplexers to support both hybrid ANN-SNN feature extraction and manifold energy evaluation as described by the mapping table. The sparsity-aware data flow shows that the sparsity-aligned decoder converts run-length-encoded input feature maps to dense data for matrix and convolution operations, and the resulting sparse outputs are re-encoded into a compact format and buffered before transferring to other cores. This sparsity-aware process sufficiently utilizes the variability of intra-layer sparsity and multi-layer data dependence to induce widely-different execution time for multiple RSPC cores. Therefore, the RSPC cores can be interconnected by globally-asynchronous fabric to achieve event-driven-based high-energy-efficient processing.

Figure 5 shows the GALS circuits for the proposed manifold-based hybrid ANN-SNN design, which addresses the limitations of both fully synchronous and fully asynchronous designs. Locally-synchronous RSPC cores with independent local clock generators (LCG) efficiently handle dense-data computations, while a globally-asynchronous fabric with handshake-based asynchronous boundary, LCG triggers, and asynchronous FIFOs enables event-driven clock triggering and dense data computation with widely-different execution time across the RSPC cores. As a result, the proposed GALS design reduces redundant clock tree power by more than 40% via shutting down the clock of RSPC/PPC core during idle state.

The 40nm GALSNP chip is evaluated on three tasks: image classification, keywords spotting (KWS), and EEG seizure detection. For MNIST and KWS tasks, the proposed design improves the average noisy-condition accuracy by 12% against the baseline. Taking EEG seizure detection as a case study, Fig. 3 compare inference and on-chip adaptation accuracy between the proposed manifold-based hybrid ANN-SNN design and gradient-based SNN baseline designs under varying SNRs. When performing EEG seizure detection, the proposed design achieves up to 24.98% higher inference accuracy at an SNR of 0.2 dB, while maintaining zero-shot on-chip adaptation with 6% inter-patient accuracy improvement without catastrophic forgetting. Fig. 6 shows that the proposed GALSNP achieves the highest energy efficiency of up to 1.01 pJ/SOP at 0.78 V and 10 MHz. Compared with the prior designs in [8]-[9], the proposed design achieves up to 97.7% and 70.9% energy efficiency improvement in inference and few-shot training, respectively, while maintaining higher accuracy. Fig. 7 presents the chip test platform, layout, measurement results, and specifications. Compared with the globally synchronous design, the proposed GALSNP achieves reduction of 43% energy consumption.

Acknowledgements:

This work is jointly supported by the NSFC (61974053 & 62274072) and the NRF, Singapore, and Cyber Security Agency of Singapore under its National Cybersecurity Research & Development Programme (NRF-NCR25-DeSNTU-0001). Z. Shen and X. Liu equally contributed to this paper. Corres. auth.: C. Wang and J. Zhou.

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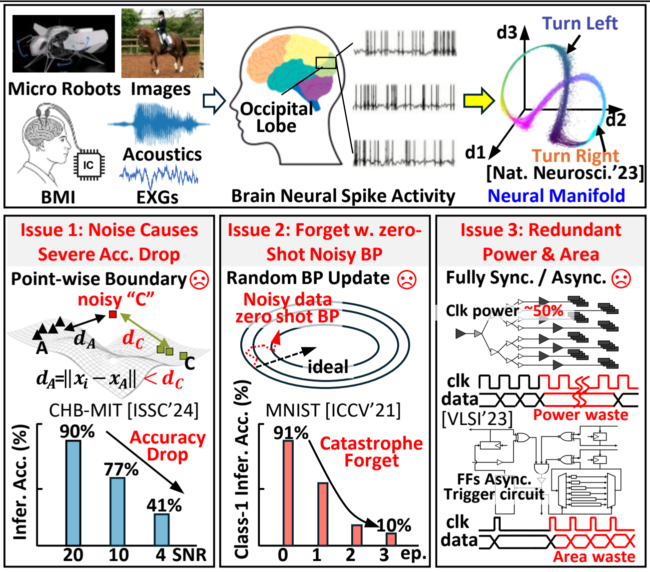


Fig. 1. SNN-based neuromorphic processors for IoT applications (top) and three existing major issues (bottom).

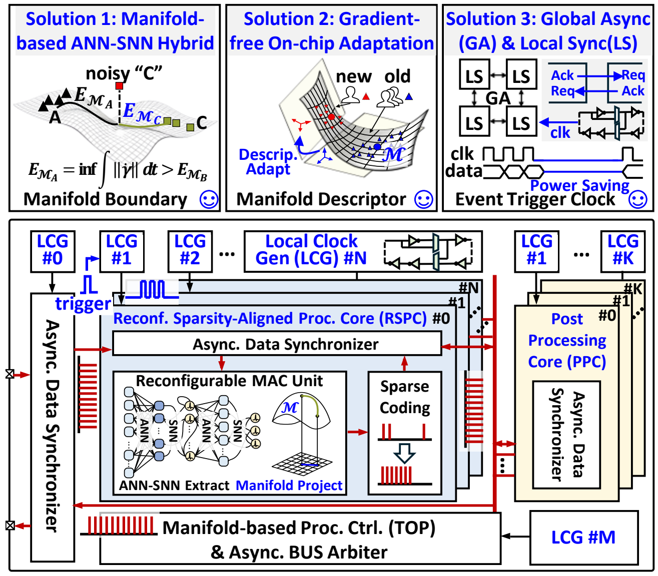


Fig. 2. Our solutions (top) and overall architecture of manifold-based GALS neuromorphic processor (bottom).

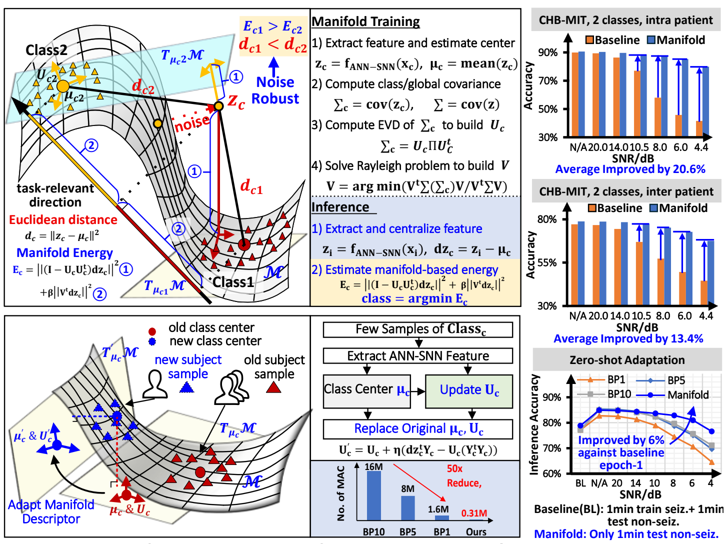


Fig. 3. Flow of proposed Manifold-based ANN-SNN hybrid noise-robust algorithm (top), gradient-free on-chip manifold descriptor adaptation (bottom) and corresponding evaluation results (right).

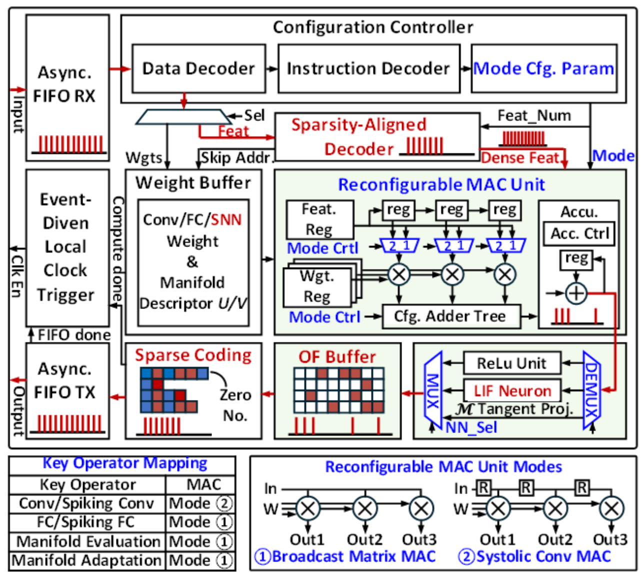


Fig. 4. Reconfigurable sparsity-aligned core and mapping table for manifold-based ANN-SNN hybrid with sparsity-aware flow.

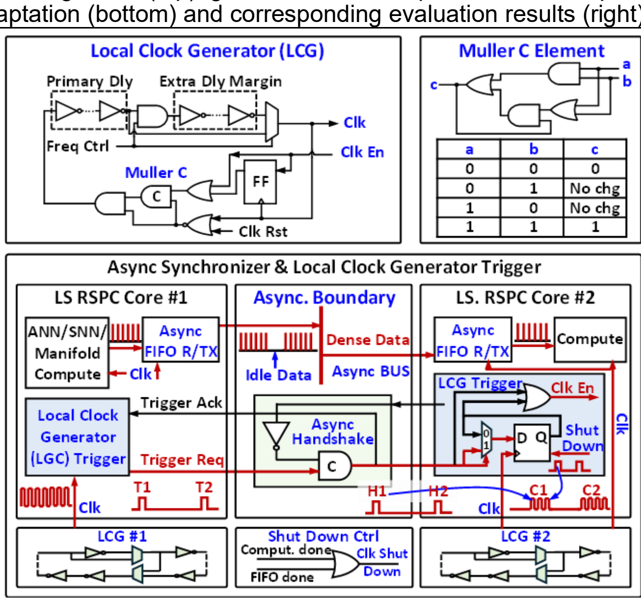
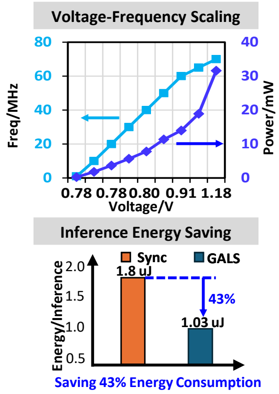


Fig. 5. Circuits of local clock generator, synchronizer and LCG trigger for globally-asynchronization among RSAC and PPC cores.

	ISSCC'22[7]	VLSI'23[4]	JSSC'24[9]	JSSC'25[8]	This Work
Tech. (nm)	28	28	28	55	40
Voltage (V)	0.5	0.55	0.56-0.9	0.78-1.2	0.78-1.2
Area (mm^2)	0.86	1.00	1.63	0.37	1.69
Architecture	Sync. DM[1]	Async. DM[1]	Async. DM[1]	Sync. Rconf.	GALS Rconf.
Freq. (MHz)	13-115	N/A	210	2.5-20	1-70
Neuron	LIF	LIF, IF	LIF, IF	M-P[2], FSN	M-P, LIF
On-chip Training Algorithm	Mod. stoch. e-prop (gradient-based)	STDP	S-TP	Back Propagation	Manifold Adaptation
Mem. Size(KB)	138	2.1	266.5	8.5	19.5
Precision	16(A) 1-16(W)	1(A) 4(W)	1(A) 8,10(W)	8(A) 8(W)	1-8(A) 8(W)
Dataset	DVS Gesture SHD[3]	FTs[4]	N-MNIST N-TIDIGIT	MNIST N-TIDIGIT	MNIST GSCD[5] CHB-MIT
Accuracy	Gest: 87.3% SHD 1word: 90.7%	FTs:97.85%	NMNIST:96% TIDIGT 1word: 92.6%	MNIST:98.3% TIDIGT 1word: 95.1%	MNIST:98.7% GSCD 2words: 98.2% CHB-MIT: 90.7%
Energy Eff. (pJ/SOP)[6]	11.3	2.09	2.91	N/A	1.01[7]
Energy Eff. (uJ per Infer/Learn Adapt.)[8]	KWS:10.7/45.0	N/A	MNIST:0.67/0.079 KWS:11.83/0.058	MNIST: 0.58/0.064 KWS:0.54/0.060	MNIST:0.39/0.023 KWS:0.27/0.050 EEG:0.24/0.049

*1: direct mapping (DM) *2: McCulloch-Pitts model *3: spiking heidelberg digits (SHD) *4: flammable and toxic gas dataset (FTs) *5: google speech command dataset (GSCD) *6: Scaled to 0.78 V *7: w/ 80% Sparsity *8: Scaled to 0.78 V, 10 MHz

Fig. 6. Comparison with state-of-the-art neuromorphic processors.



Technology	40 nm
Area	1.3 mm × 1.3 mm
Voltage	0.78V-1.2V
Frequency	1-70 MHz
Precision	1-8(A) 8(W)
Training	Manifold Adaptation
Dynamic Power	0.26-31.6 mW
Energy Efficiency	1.01-2.51 pJ/SOP
Accuracy	MNIST:98.7% GSCD: 98.2% CHB-MIT: 90.7%

Fig. 7. Chip test platform photo, chip layout and measurement results and specifications.

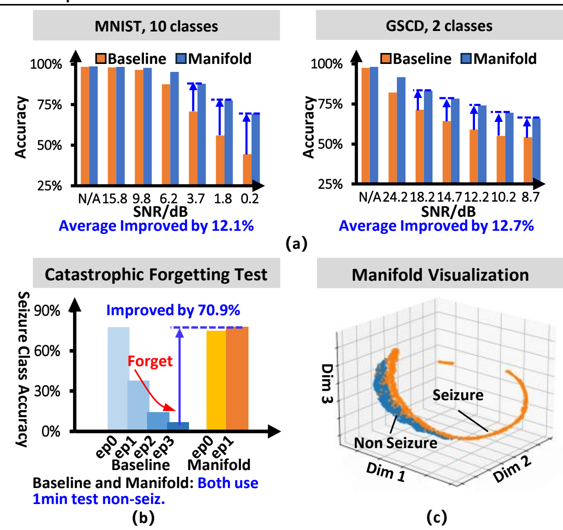


Fig. S1 (a) Inference accuracy comparison against BP-based baseline network on MNIST and GSCD dataset; (b) improved catastrophic forgetting by manifold design and (c) manifold visualization of CHB-MIT dataset.

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